Parental Housing Transfers and Life-Cycle Outcomes in Denmark

A dynamic life-cycle model of forældrekøb and intergenerational advantage

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Abstract

This paper examines how *forældrekøb* (parent-purchased properties where parents buy housing for their child) affects human capital accumulation, long-run labor market earnings, labor supply, and fertility timing in Denmark. We propose a T-period life-cycle model, where forældrekøb relaxes the budget constraint during education, shifting time from work to study and generating permanently higher wages later in life. The effect on fertility is ambiguous: higher wages make children more affordable, but also more costly in terms of foregone earnings. To identify the causal effect, we exploit the 2021 reform that made forældrekøb significantly more expensive, especially in cities like Copenhagen, creating quasi-random variation in access across otherwise similar families. We use Danish registers from Statistics Denmark to construct the forældrekøb indicator and observe life-cycle outcomes.

Research question

Does receiving a parent-purchased home during higher education causally affect educational attainment, long-run labor market income, labor supply and fertility?

Motivation

Finding affordable housing in the large cities in Denmark as a student is difficult and costly. Forældrekøb has emerged as a private solution and has gone from being a niche product to a product being actively marketed by Denmark's major banks. In 2023, 53,700 people lived in a parent-purchased home across 27,400 properties, making up 9 percent of all owner-occupied apartments in Copenhagen and Aarhus (Statistics Denmark, 2025). But it is only a solution for the lucky few. Owners of forældrekøb are heavily concentrated at the top of the income and wealth distributions, far more so than other homeowners. Students who do not have to worry about housing are expected to be more economically advantaged compared to their peers.

This motivates the question: does being able to live in a forældrekøb cause better life-cycle outcomes? Two findings motivate this question. First, if you do not have to worry about paying rent, you are more likely to work less and spend more time studying. More human capital early in life compounds into higher earnings for the rest of your career (Keane, 2011). Second, high housing costs have been found to negatively affect fertility rates across multiple countries, including Denmark (Fluchtmann et al., 2023).

If either channel holds, forældrekøb does not merely reflect existing inequality, it amplifies it. With Danish house prices having risen dramatically over the past two decades, forældrekøb has become an increasingly powerful and popular investment vehicle for wealthy parents. But if it also generates better educational and fertility outcomes for the child, it becomes a compounding intergenerational advantage operating largely outside the reach of public policy (Heckman et al., 2021).

Model framework and mechanisms

Setup

We propose a T-period life-cycle model with two phases: an education phase $t \leq T_e$ and a work phase $t > T_e$. The state variables are assets a_t, human capital k_t, children $n_t \in \{0,1\}$, and the forældrekøb indicator $\varphi \in \{0,1\}$, which is received at $t = 1$ and only affects the education phase.

Education Phase ($t \leq T_e$)

$$
\begin{aligned}
V_t(n_t, a_t, k_t; \varphi) = \max_{c_t, h_t} \quad & \frac{c_t^{1+\eta}}{1+\eta} - \beta(n_t)\frac{h_t^{1+\gamma}}{1+\gamma} + \rho\, \mathbb{E}_t[V_{t+1}] \\
\text{s.t.} \quad & a_{t+1} = (1+r)\big(a_t + w_s h_t - c_t - \bar{p}(1-\varphi\delta)\big) && (1) \\
& k_{t+1} = k_t + \theta(1-h_t) && (2)
\end{aligned}
$$

The individual splits time between work h_t and study $1-h_t$. Forældrekøb ($\varphi = 1$) reduces housing costs by a factor δ, freeing up resources that reduce the need to work during studies. More study time raises human capital k_{T_e}, which permanently raises wages in the work phase.

Work Phase ($t > T_e$)

$$
V_t(n_t, a_t, k_t) = \max_{c_t, h_t, d_t} \frac{c_t^{1+\eta}}{1+\eta} - \beta(n_t)\frac{h_t^{1+\gamma}}{1+\gamma} + \chi\, n_{t+1} + \rho\, \mathbb{E}_t\big[V_{t+1}(n_{t+1}, a_{t+1}, k_{t+1})\big] \tag{3}
$$

$$
\begin{aligned}
\text{s.t.} \quad & a_{t+1} = (1+r)\big(a_t + (1-\tau)w_t h_t - c_t\big) && (4) \\
& w_t = \bar{w}(1+\alpha k_t) && (5) \\
& k_{t+1} = k_t + h_t && (6) \\
& n_{t+1} = n_t + d_t, \qquad d_t \in \{0,1\} \text{ chosen only if } n_t = 0, \text{ else } d_t = 0 && (7)
\end{aligned}
$$

where the disutility of work depends on the presence of children: $\beta(n_t) = \beta_0 + \beta_1 n_t$ with $\beta_1 > 0$, which captures the child penalty effect as in Adda et al. (2017) and Kleven et al. (2019). And where $\eta < 0$ so period utility is CRRA with diminishing marginal utility of consumption.

As long as the household does not have a child ($n_t = 0$), it decides each period whether to have its child now. This is captured by the discrete choice $d_t \in \{0,1\}$, where $d_t = 1$ means a child arrives next period ($n_{t+1} = 1$). So before becoming parents, the household does not only decide how much to consume and work, it also decides when to become a parent. Once the child has arrived ($n_t = 1$) the choice is behind them. At that point d_t disappears and the household simply chooses consumption c_t and work hours h_t.

Human capital is accumulated in both phases but through different activities. During education only study time $(1-h_t)$ builds career-relevant human capital. Therefore the model assumes that the student job provides income but does not raise k_t. During the work phase human capital is built through labor-market experience, $k_{t+1} = k_t + h_t$, so fewer hours, for example after a birth, mean less accumulated experience and hence permanently lower wages, not just a temporary drop in the birth year. This experience channel is what turns a child into a lasting career cost, as in Adda et al. (2017).

Fertility timing is endogenous, and therefore something the household actively decides. Having the child delivers a benefit χ but raises the marginal disutility of work by β_1 from that period and onwards. As work both funds consumption and builds human capital, this captures the career cost of children and the foregone earnings from working less. The household will decide to have the child in the period where the benefits finally outweigh the costs. During the education phase fertility is fixed at $n_t = 0$, the fertility decision is therefore only active in the work phase. The model is solved by backward induction.

Forældrekøb ($\varphi = 1$) raises human capital during education and hence wages throughout the work phase, $w_t = \bar{w}(1 + \alpha k_t)$. Because forældrekøb only enters the education-phase budget, it affects fertility through these higher wages rather than directly, and the higher wages pull the timing decision in two opposing directions. Through the income effect, a higher wage and greater wealth make it easier to afford to have a child, which calls for having a child earlier. But through the substitution effect, a higher wage makes it more attractive to work more, which calls for having a child later. Which force dominates depends on the estimated parameters $\{\chi, \beta_1, \alpha\}$, so whether forældrekøb supports or delays the decision to have a child is one of the empirical questions the estimated model is designed to answer.

How the model can be used to answer the research question

We answer the research question by solving and simulating the model for two populations, $\varphi = 0$ and $\varphi = 1$, and comparing their life-cycle profiles: study time and human capital at the end of education k_{T_e}, earnings and labor supply over the work phase, and the age they decide to have their first child. The difference between the two simulated populations is the model's predicted effect of forældrekøb on each outcome.

Two caveats are worth mentioning. First, the model builds in how a forældrekøb helps, so it cannot tell us whether or not it helps having a forældrekøb. This is because it is assumed in the model that freeing up time during studies, lets you study more, which raises human capital and therefore later wages. So by construction, a forældrekøb can only raise wages, never lower them. So the model cannot be used to prove that forældrekøb matters, as we would simply get the construction back that we put in. What the model can tell us is how much a forældrekøb matters, meaning how big the wage gain is, how much it compounds over a career, and whether it makes people have children earlier or later. The model cannot tell us this in advance, as the income and substitution effects pull in opposite directions. Second, the causal effect of forældrekøb comes from the data, not the model. The 2021 reform gives us the variation we need to measure it (see Data), and we then use the model to explain that effect by splitting it into the separate channels above.

To bring the model to the data, we estimate the behavioral parameters $\{\theta, \alpha, \beta_0, \beta_1, \chi\}$ by simulated method of moments (SMM). We simulate the model for many different parameter values and keep the ones that make the simulated moments line up with the same moments computed from the Danish registers. Parameters we can observe directly are calibrated outside the model, particularly the reduced housing costs δ, which we determine from forældrekøb data as Statistics Denmark did in their analysis (see Data).

The standard parameters are fixed at conventional values from the literature: the interest rate r, the tax rate τ and the preference parameters η, γ, ρ. This lets the estimation focus on the behavioral parameters. Each of these is determined by a specific moment in the data. To determine θ, which is how productive study-time is, we look at how much people work during education for $\varphi = 1$ and $\varphi = 0$. To pin down α, which determines how much human capital affects wages, we look at how much earnings grow with age. For the child penalty β_0 and β_1 we use how much labor supply changes with age together with when the individual decides to have a child. For χ and for how strongly fertility responds, we use the age at which people decide to have their first child across $\varphi = 1$ and $\varphi = 0$.

On their own, these moments only tell us something about correlation, not what causes what. What gives them a causal interpretation is the 2021 reform. By matching how the same moments move around the reform across high- and low-exposure areas, we can isolate the effect of forældrekøb itself.

Limitations and extensions

The model has several limitations worth acknowledging. First, all individuals in the model share the same study productivity θ, meaning we cannot distinguish between forældrekøb actually causing people to study more, and recipients simply being more able to begin with, since their parents are wealthier and more educated. Forældrekøb is the only thing in the model that can make someone study more. This is why we rely on the 2021 reform for identification instead of just comparing people who had forældrekøb with those who did not.

Second, we assume that during education human capital is accumulated only through studying more, and that a student job provides income but no human capital. In reality students can have career-relevant jobs, which puts them in a better position to get better jobs after having finished their education. This means that working during studies can be beneficial for later human capital, so the model may overestimate the human capital benefit of forældrekøb, since a recipient who works less also gives up whatever human capital a relevant job would have provided.

Third, we only allow for one child, so the model speaks to the timing of the first birth, and does not take into account how many children people end up having.

Fourth, we do not take the type or length of education into account. The model captures how much people invest in human capital but not which field or degree they choose. If forældrekøb tends to shift individuals toward longer degrees, or fields of study that generate higher wages, that is an additional channel the model does not capture.

Lastly, all decisions are made by a single individual. In reality, fertility and labor supply are joint decisions made within a couple. This would be the most natural direction for an extension of the model. If fertility became a shared decision rather than an individual one, it would affect the model in different ways. The spouse's income would make the fertility timing depend on the resources of the household rather than the individual. The choice of the partner would add a new selection channel as recipients of forældrekøb are likely to match with partners with higher earnings. This would interact with the identification based on the 2021 reform and would therefore need to be handled separately. The single-agent model is kept for simplicity.

Data

Statistics Denmark conducted an analysis of parental purchases by linking register data on homeowners, business owners, residential addresses, and family relationships. The analysis was based on data from 2023 and was published in 2025 (Statistics Denmark, 2025). The analysis defines a forældrekøb as an occupied property owned by someone who does not live there, where at least one resident is a child of the owner. The forældrekøb indicator is constructed by linking the property ownership register (Ejerregistret), residential address register and CPR family relations, flagging the properties where the owner is a parent of the resident. Both privately and company-owned properties are included. We adopt the same definition and construct the forældrekøb indicator the same way. The only difference is that we build it across several years rather than only for the 2023 snapshot, so we can follow individuals over the life cycle.

For outcomes, we use educational attainment from Statistics Denmark specifically, Befolkningens højst fuldførte uddannelse, which shows the highest completed education for an individual. Earnings and labor supply is obtained from the Income Register and IDA, and fertility (births and their timing) from the population register (BEF) at Statistics Denmark. These registers are linked at the individual level through a pseudonymized CPR, which is also used to construct the forældrekøb indicator.

How far back the registers go determines what we can do. The CPR system has assigned a personal identifier to everyone since 1968. The income and labor market data in IDA and the Income Register go back to 1980. Educational attainment is available from the early 1970s. The population register (BEF) goes back to 1986. Actual hours worked are only recorded from 2008, but for earlier years we can proxy labor supply by earnings and employment. The real constraint is how far back property ownership in Ejerregistret can be reconstructed, as the forældrekøb indicator needs links between the owner, address and family. The Statistics Denmark analysis is only a snapshot from 2023, so we would need to confirm in the register documentation how far back ownership data extends.

As the reform we rely on for identification happened in 2021, the people affected by it are still young. We therefore cannot yet observe their full lifetime earnings or their completed fertility. What we can observe within this window are the outcomes that appear earlier in life, namely educational attainment, early career earnings and the timing of the first birth, which happens in the 20s and 30s. Observing the full life cycle would require following these cohorts for longer.

Identification

The main challenge is that forældrekøb is not handed out randomly. The people who get it have wealthier and more educated parents, and they are probably more able to begin with. So if we simply compare people who had forældrekøb with people who did not, we cannot tell whether their better outcomes come from the forældrekøb itself or from the advantages they already had. This is the central identification problem of the paper.

We deal with this using the 2021 reform, which made forældrekøb much more expensive, especially in larger cities like Copenhagen where market prices were far above the public assessment the old rules were based on. After the reform, some families could no longer afford forældrekøb for reasons that have nothing to do with their child's ability. We use this change in access to isolate the effect of forældrekøb itself. By looking at how outcomes change for the families the reform pushed out, compared with similar families it did not affect, we can separate the effect of forældrekøb from the advantages these families already had.

In the estimation this is also what identifies the causal parameters. We ask the model to match how outcomes like study time, earnings and age when getting a child move around the reform, instead of just matching the raw difference between forældrekøb and non-forældrekøb individuals, which would still be mixed up with selection.

References

Adda, J., Dustmann, C., & Stevens, K. (2017). The career costs of children. *Journal of Political Economy*, 125(2), 293–337.

Statistics Denmark (2025). Forældrekøbte boliger - hvem er beboerne, og hvem er forældrene? *Analyse*, 4. april 2025.

Fluchtmann, J., van Veen, V., & Adema, W. (2023). Fertility, employment and family policy: A cross-country panel analysis. *OECD Social, Employment and Migration Working Papers*, No. 299.

James J. Heckman & Rasmus Landersø (2021). Lessons from Denmark about Inequality and Social Mobility. *Labour Economics*.

Keane, M.P. (2011). Labor supply and taxes: A survey. *Journal of Economic Literature*, 49(4), 961–1075.

Kleven, H., Landais, C., & Søgaard, J.E. (2019). Children and gender inequality: Evidence from Denmark. *American Economic Journal: Applied Economics*, 11(4), 181–209.